BSc 2023/2024

For internal students of Imperial College.

Mathematics Foundations (BUSI40001)

2 hours

Sample Exam, Dec 2023

Instructions for the exam

- For Section A, internet access with ChatGPT is allowed and a computer with R or Python is needed.
- Section B is closed-book and paper-based. Section B is to made accessible to the students after completing Section A.
- Answer ALL questions (Please note that not all questions carry equal marks, the mark breakdown is indicated throughout the exam).
- It is important to write neatly and clearly.
- If a question or part of a question is unclear to you, state any assumptions you need in order to complete the question.

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SECTION A

You have 45 minutes for Section A. Answer all questions. You must close your laptop after 45 minutes.

1. (23 marks) The XL file

`SampleExamData_2023.xlsx`

contains data for annual returns on the S&P 500 Index and Berkshire Hathaway.

This question is about using OLS linear regression to analyze the relationship between returns for Berkshire Hathaway and returns on the S&P 500 Index.

(a) Write and use Python or R code to read the data from the XL file and create a column vector $\underline{x}$ containing returns for the S&P 500 Index and a column vector of the same size, $\underline{y}$, containing returns for Berkshire Hathaway. [4 marks]

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Load the data from the Excel file
excel_file_path = 'SampleExamData_2023.xlsx'
data = pd.read_excel(excel_file_path)

# Prepare the independent variable X
X = data['SP500'].values

# Prepare the dependent variable Y
Y = data['Berkshire'].values
```

(b) Write out the linear regression of Berkshire Hathaway returns against S & P 500 returns in the matrix form

$$\underline{y} = \mathbf{X}\underline{\beta} + \underline{\epsilon},$$

where $\mathbf{X}$, $\underline{\beta}$, and $\underline{\epsilon}$ are to be defined. [2 marks]

$\underline{e} = (1, \ldots, 1)^\top$ is $N \times 1$ vector of ones, where N is the number of observations.

$$\mathbf{X} = [\underline{e}, \underline{x}]$$

is the N by 2 design matrix,

$$\underline{\beta} = [\beta_1, \beta_2]^\top$$

is the 1 by 2 parameter vector, and $\underline{\epsilon}$ is $N \times 1$ error vector.

(c) Show that the expression for $\underline{\beta}$ which minimizes $\underline{\epsilon}^\top \underline{\epsilon}$ is given by

$$\widehat{\underline{\beta}} = \left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\underline{y}.$$

[5 marks]

Observe that $\underline{\epsilon}^\top \underline{\epsilon}$, where $\underline{\epsilon} = \underline{y} - \mathbf{X}\underline{\beta}$, is minimized by projecting $\underline{y}$ orthogonally into the column space of $\mathbf{X}$. Therefore, we want to choose $\underline{\beta}$ such that $\mathbf{X}^\top(\underline{y} - \mathbf{X}\underline{\beta}) = \underline{0}$. We shall denote the parameter vector which satisfies the latter equation via $\widehat{\underline{\beta}}$ and so

$$\mathbf{X}^\top(\underline{y} - \mathbf{X}\widehat{\underline{\beta}}) = \underline{0}$$

Therefore,

$$\mathbf{X}^\top \underline{y} = \mathbf{X}^\top \mathbf{X} \underline{\widehat{\beta}}.$$

Now $\mathbf{X}^\top \mathbf{X}$ is a 2 by 2 square matrix. Now

$$\underline{\widehat{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \underline{y}.$$

(d) Show that the fitted values are given by

$$\underline{\hat{y}} = \mathbf{H}\underline{y}$$

where

$$\mathbf{H} \equiv \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T$$

[3 marks]

The fitted values of the dependent variable are given by $\underline{\hat{y}} = \underline{\widehat{\beta}}\mathbf{X}$. Therefore,

$$\underline{\hat{y}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \underline{y},$$

and so

$$\underline{\hat{y}} = \mathbf{H}\underline{y},$$

where

$$\mathbf{H} \equiv \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T.$$

(e) Show that $\mathbf{H}$ is idempotent and symmetric. [4 marks]

$$\begin{aligned}
\mathbf{H}^2 &= \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T \\
&= \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T \\
&= \mathbf{H},
\end{aligned}$$

and so $\mathbf{H}$ is idempotent. Now,

$$\begin{aligned}
\mathbf{H}^\top &= [\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T]^\top \\
&= [\mathbf{X}^\top]^\top[\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top \\
&= \mathbf{X}[\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top \\
&= \mathbf{X}[\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top\mathbf{X}^\top \\
&= \mathbf{X}[\left(\mathbf{X}^T\mathbf{X}\right)^\top]^{-1}\mathbf{X}^\top \\
&= \mathbf{X}\left[\left(\mathbf{X}^\top(\mathbf{X}^T)^\top\right)\right]^{-1}\mathbf{X}^\top \\
&= \mathbf{X}\left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1}\mathbf{X}^\top \\
&= \mathbf{H},
\end{aligned}$$

and so $\mathbf{H}$ is symmetric.

(f) Write and use R or Python code to find $\underline{\widehat{\beta}}$. [5 marks]

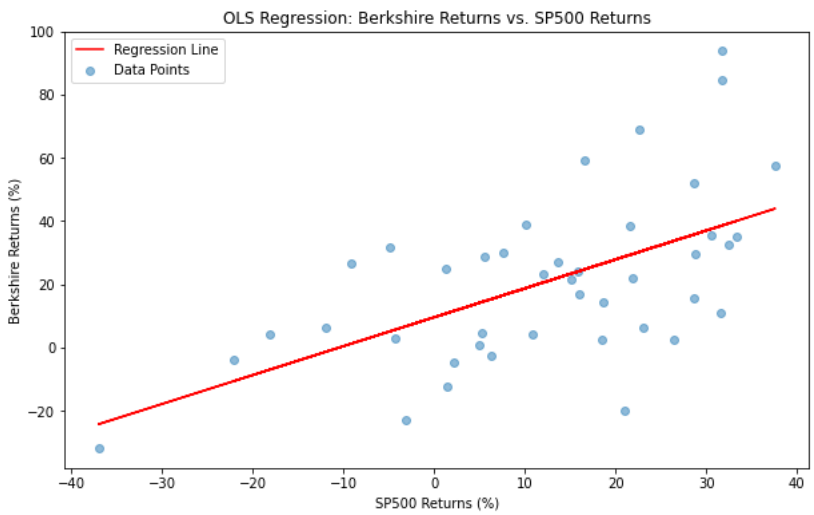
Extend the existing code as follows:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Load the data from the Excel file
excel_file_path = 'SampleExamData_2023.xlsx'
data = pd.read_excel(excel_file_path)

# Prepare the independent variable X by adding a constant (intercept term)
X = data['SP500'].values
X_matrix = np.vstack([np.ones(len(X)), X]).T  # Stack a column of ones and the X values for the intercept
                                                term

# Prepare the dependent variable Y
Y = data['Berkshire'].values

# Calculate the beta_hat vector (OLS estimator) using the normal equation
beta_hat = np.linalg.inv(X_matrix.T @ X_matrix) @ X_matrix.T @ Y

#NOT NEEDED FOR SOLUTION
# Plotting the data points
plt.figure(figsize=(10, 6))
plt.scatter(X, Y, alpha=0.5, label='Data Points')

# Computing the regression line values
regression_line = X_matrix @ beta_hat

# Plotting the regression line
plt.plot(X, regression_line, color='red', label='Regression Line')

# Adding title and labels to the plot
plt.title('OLS Regression: Berkshire Returns vs. SP500 Returns')
plt.xlabel('SP500 Returns (%)')
plt.ylabel('Berkshire Returns (%)')
plt.legend()

# Display the plot
plt.show()
```

Using the code, we obtain

```
array([9.58661776, 0.91421521])
```

Hence, $\widehat{\underline{\beta}} = [9.6, 0.9]^\top$, to 1 d.p.



The code does not need to make the plot, but it is useful for understanding.

2. (a) Find and classify all critical points (local maximum, local minimum or saddle point) of

$$(x+y)^2 - 3xy^2.$$

You may write code in R or Python to carry out your calculations and draw any plots. [12 marks]

You need to know how to use ChatGPT to generate this code. What will your initial prompt be?

```python
import sympy as sp
import matplotlib.pyplot as plt
import numpy as np
from mpl_toolkits.mplot3d import Axes3D

# Define the symbols
x, y = sp.symbols('x y')

# Define the function
f = (x+y)**2 - 3*x*(y**2)

# Compute the first derivatives (gradient)
grad_f = [sp.diff(f, var) for var in (x, y)]

# Solve for critical points
critical_points = sp.solve(grad_f, (x, y))

# Ensure critical points are in the correct format
if isinstance(critical_points, dict):
    critical_points = [tuple(critical_points.values())]
elif isinstance(critical_points, tuple):
    critical_points = [critical_points]

# Compute the Hessian matrix
H = sp.hessian(f, (x, y))

# Function to determine the nature of critical points
def nature_of_critical_point(Hessian, point):
    H_sub = Hessian.subs({x: point[0], y: point[1]})
    eigenvals = H_sub.eigenvals()
    if all(val > 0 for val in eigenvals):
        return "Local Minimum"
    elif all(val < 0 for val in eigenvals):
        return "Local Maximum"
    else:
        return "Saddle Point"

# Analyze each critical point
results = [(point, nature_of_critical_point(H, point)) for point in critical_points]


#NOT NEEDED FOR SOLUTION
# Define the function for plotting
def f_plot(x, y):
    return (x+y)**2 - 3*x*(y**2)

# Create a grid of x and y values for plotting
x_vals = np.linspace(-200, 200, 400)
y_vals = np.linspace(-8, 8, 400)
x_vals, y_vals = np.meshgrid(x_vals, y_vals)

# Calculate z values for the grid
z_vals = f_plot(x_vals, y_vals)

# Create plots
fig = plt.figure(figsize=(12, 6))

# 3D surface plot
ax1 = fig.add_subplot(121, projection='3d')
ax1.plot_surface(x_vals, y_vals, z_vals, cmap='viridis', edgecolor='none')
ax1.set_title('3D Surface Plot')
ax1.set_xlabel('X axis')
ax1.set_ylabel('Y axis')
ax1.set_zlabel('Z axis')

# Contour plot
ax2 = fig.add_subplot(122)
contour = ax2.contourf(x_vals, y_vals, z_vals, cmap='viridis')

# Mark the critical points on the contour plot
for point, nature in results:
    ax2.plot(*point, 'ro') # Mark each critical point

ax2.set_title('Contour Plot with Critical Points')
ax2.set_xlabel('X axis')
ax2.set_ylabel('Y axis')

# Add a color bar for the contour plot
fig.colorbar(contour, ax=ax2, orientation='vertical')

# Show the plot
plt.show()
```

```
results
```

Results output:

```
[((0, 0), 'Saddle Point'), ((1/2, 1), 'Saddle Point')]
```

(b) Find the local maxima and minima of $(x+y)^2 - 4y$ on the circle $x^2 + y^2 - 16 = 0$. You may use Python or R to perform calculations. [8 marks]

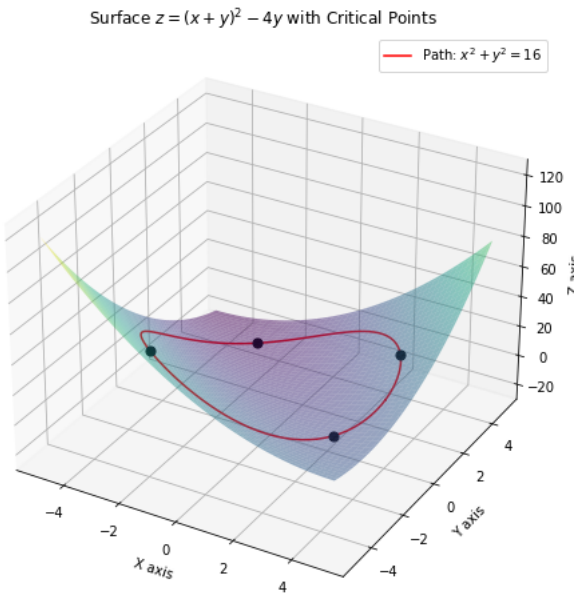
We could use Lagrange multipliers. However, we can also simplify the problem by using polar coordinates which include the constraint: $x = 4\cos\theta$, $y = 4\sin\theta$, where $0 \leq \theta < 2\pi$. Hence,

$$\begin{aligned}
(x+y)^2 - 4y &= 16\left[(\cos\theta + \sin\theta)^2 - \sin\theta\right] \\
&= 16\left[\cos^2\theta + \sin^2\theta + 2\cos\theta\sin\theta - \sin\theta\right] \\
&= 16\left[1 + 2\cos\theta\sin\theta - \sin\theta\right] \\
&= 16\left[1 + \sin 2\theta - \sin\theta\right] \\
&= 16\left[1 + h(\theta)\right]
\end{aligned}$$

where $h(\theta) = \sin 2\theta - \sin\theta$. Therefore the constrained problem reduces to choosing θ to find the local maxima and minima of $h(\theta)$, where $0 \leq \theta < 2\pi$. We can use Python to solve this problem numerically.

```
import sympy as sp
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Define the symbol
theta = sp.symbols('theta')

# Define the function h(theta)
h = sp.sin(2*theta) - sp.sin(theta)

# Compute the first derivative
h_prime = sp.diff(h, theta)

# Find critical points by solving h'(theta) = 0
critical_points = sp.solveset(h_prime, theta, domain=sp.Interval(0, 2*sp.pi))

# Compute the second derivative
h_double_prime = sp.diff(h_prime, theta)

# Function to classify the critical points
def classify_point(point):
    second_derivative = h_double_prime.subs(theta, point)
    if second_derivative > 0:
        return "Local Minimum"
    elif second_derivative < 0:
        return "Local Maximum"
    else:
        return "Inconclusive"

# Classify each critical point and calculate numerical coordinates
extrema = []
for point in critical_points:
    nature = classify_point(point)
    numerical_coords = (4*sp.cos(point).evalf(), 4*sp.sin(point).evalf())
    rounded_coords = (round(numerical_coords[0], 2), round(numerical_coords[1], 2))
    z_coord = (rounded_coords[0] + rounded_coords[1])**2 - 4*rounded_coords[1]
    extrema.append((point.evalf(), nature, rounded_coords, z_coord))


#NOT NEEDED FOR SOLUTION
# Plotting
fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')

# Plot the surface
x = np.linspace(-5, 5, 100)
y = np.linspace(-5, 5, 100)
x, y = np.meshgrid(x, y)
z = (x + y)**2 - 4*y
ax.plot_surface(x, y, z, alpha=0.5, cmap='viridis')
```

```python
# Plot the circle on the surface
theta_vals = np.linspace(0, 2*np.pi, 100)
x_circle = 4 * np.cos(theta_vals)
y_circle = 4 * np.sin(theta_vals)
z_circle = (x_circle + y_circle)**2 - 4*y_circle
ax.plot(x_circle, y_circle, z_circle, color='red', label='Path: $x^2 + y^2 = 16$')

# Plot the critical points
for _, _, coords, z_coord in extrema:
    ax.scatter(*coords, z_coord, color='black', s=50)

ax.set_xlabel('X axis')
ax.set_ylabel('Y axis')
ax.set_zlabel('Z axis')
ax.set_title('Surface $z = (x + y)^2 - 4y$ with Critical Points')
ax.legend()

plt.show()
```



The code does not need to make the plot, but it is useful for understanding.
Running the code gives:
θ (numerical) = 0.567829372928644, Type: Local Maximum, Coords (numerical, rounded) = (3.37, 2.15)
θ (numerical) = 2.20566319792847, Type: Local Minimum, Coords (numerical, rounded) = (-2.37, 3.22)
θ (numerical) = 4.07752210925112, Type: Local Maximum, Coords (numerical, rounded) = (-2.37, -3.22)
θ (numerical) = 5.71535593425094, Type: Local Minimum, Coords (numerical, rounded) = (3.37, -2.15)

SECTION B

You have 1 hour 15 minutes for Section B. Answer all questions.

1. (34 marks) Consider a real sequence $(a_n)_n$

(a) What does it mean to say that a sequence of real numbers $(a_n)_n$ converges to $\ell \in \mathbb{R}$? [2 marks]

This means that for every strictly positive number ϵ, there exists at least one natural number N such that for all $n \geq N$, the distance between a_n and ℓ is less than ϵ, i.e.

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)\, |a_n - \ell| < \epsilon.$$

(b) What does it mean to say that a sequence of real numbers $(a_n)_n$ is bounded? [1 mark]

A sequence (a_n) is bounded if $(\exists L, U \in \mathbb{R})(\forall n \in \mathbb{N}) L \leq a_n \leq U$.

(c) Prove that every convergent sequence is bounded. [6 marks]

We want to show that

$$\begin{aligned}&(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)|a_n - a| < \epsilon \\ &\Rightarrow (\exists L, U \in \mathbb{R})(\forall n \in \mathbb{N}) L \leq a_n \leq U.\end{aligned}$$

We start with a convergent sequence $(a_n)_n$. Using the definition of convergence, we have

$$\begin{aligned}&(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)|a_n - a| \\ \iff &(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N) a - \epsilon < a_n < a + \epsilon\end{aligned}$$

Let us set $\epsilon = 1$. Denote the associated N by N_1

$$(\forall n \geq N_1) a - 1 < a_n < a + 1.$$

Now for $n < N_1$ we need to consider $\{a_1, \ldots, a_{N_1 - 1}\}$, which will have a maximum element, denoted by M and a minimum element, denoted by m. Therefore,

$$(\forall n \leq N_1 - 1) m \leq a_n \leq M.$$

Hence,

$$\begin{aligned}&(\forall n \geq N_1) a - 1 < a_n < a + 1 \wedge (\exists m, M \in \mathbb{R})(\forall n \leq N_1 - 1) m \leq a_n \leq M \\ \Rightarrow &(\forall n \in \mathbb{N})(\exists m, M \in \mathbb{R}) \min\{a - 1, m\} \leq a_n \leq \max\{a + 1, M\} \\ \Rightarrow &(\forall n \in \mathbb{N})(\exists L, U \in \mathbb{R}) L \leq a_n \leq U.\end{aligned}$$

(d) Show that if $0 < c < 1$, then the sequence $(c^n)_n$ converges to 0. [4 marks]

We want to show that

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)|c^n| < \epsilon$$

Now, since $0 < c < 1$, $|c^n| = c^n$, so we want show that

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N) c^n < \epsilon$$

We consider the inequality $c^n < \epsilon$ and note that

$$c^n < \epsilon \iff n \ln c < \ln \epsilon \iff n > \frac{\ln \epsilon}{\ln c}.$$

By the Archimedean Property

$$\begin{aligned}&(\forall\epsilon>0)(\exists N\in\mathbb{N})N>\frac{\ln\epsilon}{\ln c}\\ \iff&(\forall\epsilon>0)(\exists N\in\mathbb{N})c^N<\epsilon.\end{aligned}$$

Using the fact that $0<c<1$, we know that $c^{N+1}=c\times c^N<c^N$. Now

$$(\forall n\geq N)c^n=c^{n-N}c^N\leq c^N.$$

Therefore,

$$0<c<1\Rightarrow(\forall\epsilon>0)(\exists N\in\mathbb{N})(\forall n\geq N)c^n<\epsilon\iff c^n\to 0$$

(e) Let $(a_n)_n,(x_n)_n,(b_n)_n$ be sequences of real numbers. Suppose that for all $n\in\mathbb{N}$, $a_n\leqslant x_n\leqslant b_n$ and $(a_n)_n,(b_n)_n$ converge to $L\in\mathbb{R}$. Prove that $(x_n)_n$ converges to L too. [5 marks]

We want to prove that

$$((\forall n\in\mathbb{N})\,a_n\leq x_n\leq b_n)\wedge a_n\to L\wedge b_n\to L\Rightarrow x_n\to L.$$

We know that

$$\begin{aligned}&(\forall\epsilon>0)(\exists N_a\in\mathbb{N})(\forall n\geq N_a)|a_n-L|<\epsilon\\ &\wedge(\forall\epsilon>0)(\exists N_b\in\mathbb{N})(\forall n\geq N_b)|b_n-L|<\epsilon\\ \iff&(\forall\epsilon>0)(\exists N_a\in\mathbb{N})(\forall n\geq N_a)L-\epsilon<a_n<L+\epsilon\\ &\wedge(\forall\epsilon>0)(\exists N_b\in\mathbb{N})(\forall n\geq N_b)L-\epsilon<b_n<L+\epsilon\end{aligned}$$

Setting $N=\max\{N_a,N_b\}$, and noting that $(\forall n\in\mathbb{N})\,a_n\leq x_n\leq b_n$ we see that

$$\begin{aligned}&(\forall\epsilon>0)(\exists N\in\mathbb{N})(\forall n\geq N)L-\epsilon<a_n\leq x_n\leq b_n<L+\epsilon\\ \Rightarrow&(\forall\epsilon>0)(\exists N\in\mathbb{N})(\forall n\geq N)L-\epsilon<x_n<L+\epsilon\\ \iff&(\forall\epsilon>0)(\exists N\in\mathbb{N})(\forall n\geq N)|x_n-L|<\epsilon\\ \iff&x_n\to L.\end{aligned}$$

(f) Let $(a_n)_n$ be a sequence of positive real numbers. For the following statements provide either a proof or a counterexample.

i. If $\frac{a_{n+1}}{a_n}\to 1$ then (a_n) converges. [6 marks]

Consider the sequence $a_n=n$. We shall show that $\frac{a_{n+1}}{a_n}\to 1$, but that this sequence does not converge.
To show no convergence, by definition, we must show that

$$(\nexists L\in\mathbb{R})(\forall\epsilon>0)(\exists N\in\mathbb{N})(\forall n\geq N)|n-L|<\epsilon.$$

We shall prove this by contradiction. Suppose

$$(\exists L\in\mathbb{R})(\forall\epsilon>0)(\exists N\in\mathbb{N})(\forall n\geq N)|n-L|<\epsilon.$$

Now consider $|N+1-L|<\epsilon\iff L-\epsilon<N<L+\epsilon$. If we choose $\epsilon=1/2$ (any ϵ strictly between 0 and 1 will do!), then $N+1>L+\epsilon$, which implies $|N+1-L|>\epsilon$. [Draw a picture of the real line illustrating $|N+1-L|<\epsilon$ for $\epsilon=1/2$ if you want to understand this easily!] Therefore, we have a found an integer greater than or

equal to N such that $|n - L| > \epsilon = 1/2$, contradicting our assumption. Therefore $a_n = n$ does not converge.
We now consider $b_n = \frac{a_{n+1}}{a_n} = \frac{n+1}{n} = 1 + \frac{1}{n}$. Consider the inequality $|b_n - 1| < \epsilon$

$$|b_n - 1| < \epsilon \iff \frac{1}{n} < \epsilon \iff n > \frac{1}{\epsilon}$$

By the Archimedean Property $(\forall \epsilon > 0)(\exists N \in \mathbb{N}) N > \frac{1}{\epsilon}$. Therefore,

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N) n > \frac{1}{\epsilon}$$

Hence,

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N) \left| \frac{n+1}{n} - 1 < \epsilon \right|,$$

i.e. $\frac{a_{n+1}}{a_n} = \frac{n+1}{n}$ converges to 1.
We have thus found a counterexample to the given statement, thereby showing it is false.

ii. If $(a_n)_n$ converges then $\frac{a_{n+1}}{a_n} \to 1$. [10 marks]

From the Algebra of Limits, we know if a_n converges to a non-zero limit, then the above statement will be true. We shall therefore consider sequences which converge to zero, with different rates of convergence and use them to construct a new sequence, the ratio of which does not converge.
We can show that $\frac{1}{n}$ and $\frac{1}{n^2}$ converge to zero, but $\frac{1/n}{1/n^2} = n$ which does not converge.
We therefore define

$$a_n = \begin{cases} \frac{1}{n} & \text{if } n \text{ is even} \\ \frac{1}{n^2} & \text{if } n \text{ is odd} \end{cases}$$

Now

$$(\forall n \in \mathbb{N}) |a_n| = a_n \leq \frac{1}{n} \tag{1}$$

By the Archimedean Property

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N}) N > \frac{1}{\epsilon} \iff (\forall \epsilon > 0)(\exists N \in \mathbb{N}) \frac{1}{N} < \epsilon$$

Therefore,

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N) \frac{1}{n} < \epsilon.$$

Hence, using (1), we see that

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N) |a_n| < \epsilon,$$

so we have proved that (a_n) converges to 1.
We now consider the sequence (b_n), where $b_n = \frac{a_{n+1}}{a_n}$.

$$b_n = \begin{cases} \frac{n}{(n+1)^2} & \text{if } n \text{ is even} \\ \frac{n^2}{n+1} & \text{if } n \text{ is odd} \end{cases}$$

Now we shall show the sequence (b_n) is unbounded. We have already shown that every convergent sequence is bounded, which is equivalent to saying that an unbounded sequences does not converge. Therefore, non-convergence follows from unboundedness.
By the Archimedean Property,

$$(\forall M > 0)(\exists n \in \mathbb{N}) n > M,$$

and so

$$(\forall M > 0)(\exists n \in \mathbb{N}) \frac{n}{2} > M.$$

Therefore the sequence $(n/2)_n$ is unbounded. We note that $\frac{n^2}{n+1} > \frac{1}{2}n$. Therefore, $\forall M > 0$, we can choose an odd n to ensure $b_n > M$, and so

$$(\forall M > 0)(\exists n \in \mathbb{N}) b_n > M,$$

which means (b_n) is unbounded.
We have therefore shown the given statement is false via constructing a counterexample.

2. (33 marks)

(a) Find the orthogonal projection of the vector $\underline{b} = (1, 2, 6)^\top$ onto the plane $x + y + z = 0$ in $\mathbb{R}^3$, where the inner product is the standard scalar product. [4 marks]

The plane $x + y + z = 0$ is a 2-d subspace of $\mathbb{R}^3$ and we shall denote this 2-d subspace by U.Now

$$U = \{(-(\lambda + \mu), \lambda, \mu)^\top \in \mathbb{R}^3 : \lambda, \mu \in \mathbb{R}\} = \{\lambda(-1, 1, 0)^\top + \mu(-1, 0, 1)^\top : \lambda, \mu \in \mathbb{R}\}$$

To perform the orthogonal projection, we shall use an orthnormal basis for U. We can set the first element of this basis equal to $\frac{1}{\sqrt{2}}(1, -1, 0)^\top \in U$. We shall find the second element by looking for μ such that

$$\left((1, -1, 0)^\top + \mu(-1, 0, 1)^\top\right) \cdot (1, -1, 0)^\top = 0.$$

Hence,

$$2 + \mu(-1) = 0 \iff \mu = 2.$$

We therefore obtain the vector $(-1, -1, 2)^\top$, which is orthogonal to $\frac{1}{\sqrt{2}}(1, -1, 0)^\top$. Hence,

$$\left\{\frac{1}{\sqrt{2}}(1, -1, 0)^\top, \frac{1}{\sqrt{6}}(1, 1, -2)^\top\right\}$$

is an orthonormal basis for U. For ease of notation, define

$$\underline{a}_1 = \frac{1}{\sqrt{2}}(1, -1, 0)^\top$$
$$\underline{a}_2 = \frac{1}{\sqrt{6}}(1, 1, -2)^\top.$$

Hence, the orthogonal of projection of $\underline{b}$ onto U is given by

$$\begin{aligned}\sum_{i=1}^{2}(\underline{b}\cdot\underline{a}_i)\underline{a}_i &= ((1,2,6)^\top\cdot\frac{1}{\sqrt{2}}(1,-1,0)^\top)\frac{1}{\sqrt{2}}(1,-1,0)^\top \\ &+((1,2,6)^\top\cdot\frac{1}{\sqrt{6}}(1,1,-2)^\top)\frac{1}{\sqrt{6}}(1,1,-2)^\top \\ &=(-2,-1,3)^\top.\end{aligned}$$

(b) Let V be the vector space of real polynomials of degree $\leqslant 2$ and define the inner product as follows

$$\begin{aligned}\langle\cdot,\cdot\rangle : V\times V &\longrightarrow \mathbb{R} \\ (f,g) &\longmapsto \frac{1}{3}\sum_{k\in\{-1,0,1\}} f(k)\cdot g(k)\end{aligned}$$

i. Let U be the subspace of V spanned by 1 and x.

A. Show that 1 and x are orthogonal and use this to find an orthonormal basis for U. [5 marks]

We first show 1 and x are linearly independent by considering the equation

$$(\forall x\in\mathbb{R})\mu+\lambda x=0.$$

It follows that $\mu=\lambda=0$, and so 1 and x are linearly independent. $\{1,x\}$ is a spanning set of U by construction, and with linear independence, we see that $\{1,x\}$ is a basis of U.
Applying the definition for the inner product,

$$\langle 1,x\rangle=\frac{1}{3}\left[(1)(-1)+(1)(0)+(1)(1)\right]=0$$

Therefore, 1 and x are orthogonal, so $\{1,x\}$ is an orthogonal basis of U. We now normalize 1 and x to obtain an orthonormal basis of U.

$$\|1\|^2=\langle 1,1\rangle=\frac{1}{3}\left[(1)(1)+(1)(1)+(1)(1)\right]=1\Rightarrow\|1\|=1$$

$$\|x\|^2=\langle x,x\rangle=\frac{1}{3}\left[(1)(1)+(0)(0)+(-1)(-1)\right]=2/3\Rightarrow\|x\|=\sqrt{\frac{2}{3}}.$$

Therefore, $\{1,\sqrt{\frac{3}{2}}x\}$ is an orthonormal basis of U.

B. Find a set which serves as an orthonormal basis for V and then prove it is indeed a basis of V. [10 marks]

$f(x)=1, g(x)=\sqrt{\frac{3}{2}}x$ is an orthonormal basis for U
Define

$$h(x)=x^2-\left\langle f,x^2\right\rangle f(x)-\left\langle g,x^2\right\rangle g(x)$$

All I am doing is substracting the orthogonal projection of x^2 into U from x^2. This resulting vector will be orthogonal to all elements of U and I can then normalize it.
We now show that $\langle h,f\rangle=\langle h,g\rangle=0$.

$$\begin{aligned}
\langle h, f\rangle &= \left\langle x^2, f\right\rangle - \left\langle f, x^2\right\rangle \langle f, f\rangle - \left\langle g, x^2\right\rangle \langle g, f\rangle \\
&= \left\langle x^2, f\right\rangle - \left\langle f, x^2\right\rangle \times 1 - \left\langle g, x^2\right\rangle \times 0 \\
&= \left\langle x^2, f\right\rangle - \left\langle f, x^2\right\rangle \\
&= \left\langle x^2, f\right\rangle - \left\langle x^2, f\right\rangle \\
&= 0.
\end{aligned}$$

$$\begin{aligned}
\langle h, g\rangle &= \left\langle x^2, g\right\rangle - \left\langle f, x^2\right\rangle \langle f, g\rangle - \left\langle g, x^2\right\rangle \langle g, g\rangle \\
&= \left\langle x^2, g\right\rangle - \left\langle f, x^2\right\rangle \times 0 - \left\langle g, x^2\right\rangle \times 1 \\
&= \left\langle x^2, g\right\rangle - \left\langle g, x^2\right\rangle \\
&= \left\langle x^2, g\right\rangle - \left\langle x^2, g\right\rangle \\
&= 0.
\end{aligned}$$

Now

$$\begin{aligned}
\left\langle f, x^2\right\rangle &= \left\langle 1, x^2\right\rangle \\
&= \frac{1}{3}[(1)(1) + (1)(0) + (1)(1)] = 2/3
\end{aligned}$$

and

$$\begin{aligned}
\left\langle g, x^2\right\rangle &= \left\langle x, x^2\right\rangle \\
&= \sqrt{\frac{3}{2}}\frac{1}{3}[(-1)(1) + (0)(0) + (1)(1)] = 0
\end{aligned}$$

We thus have the vector

$$h(x) = x^2 - \frac{2}{3}.$$

Now

$$\|h\|^2 = \langle h, h\rangle = \frac{1}{3}\left[\left(1 - \frac{2}{3}\right)^2 + \left(-\frac{2}{3}\right)^2 + \left(1 - \frac{2}{3}\right)^2\right] = \frac{2}{9}$$

We therefore define

$$j(x) = \frac{3}{\sqrt{2}}h(x) = \frac{3}{\sqrt{2}}\left(x^2 - \frac{2}{3}\right),$$

so that $\|j\| = 1$.

We have shown that each vector in the set $S = \{1, \sqrt{\frac{3}{2}}x, \frac{3}{\sqrt{2}}\left(x^2 - \frac{2}{3}\right)\} = \{f(x), g(x), j(x)\}$ has norm 1 and each pair is mutually orthogonal. If we can show that $\{1, \sqrt{\frac{3}{2}}x, \frac{3}{\sqrt{2}}\left(x^2 - \frac{2}{3}\right)\}$ is basis for V, it will be an orthonormal basis. We consider the equation

$$(\forall x \in \mathbb{R}) a_1 f(x) + a_2 g(x) + a_3 j(x) = 0.$$

Exploiting pairwise orthonormality, we see that for all $x \in \mathbb{R}$

$$\begin{aligned}\langle a_1 f(x) + a_2 g(x) + a_3 j(x), f(x)\rangle &= a_1\\ \langle a_1 f(x) + a_2 g(x) + a_3 j(x), g(x)\rangle &= a_2\\ \langle a_1 f(x) + a_2 g(x) + a_3 j(x), j(x)\rangle &= a_3\end{aligned}$$

It follows that $a_1 = a_2 = a_3 = 0$, so S is a linearly independent set. We now show S spans V. Consider $ax^2 + bx + c$. We know that

$$\begin{aligned}x^2 &= \frac{\sqrt{2}}{3} j(x) + \frac{2}{3} f(x)\\ x &= \sqrt{\frac{2}{3}} g(x)\end{aligned}$$

Therefore,

$$\begin{aligned}ax^2 + bx + c &= a\left[\frac{\sqrt{2}}{3} j(x) + \frac{2}{3} f(x)\right] + b\sqrt{\frac{2}{3}} g(x) + cf(x)\\ &= a\frac{\sqrt{2}}{3} j(x) + b\sqrt{\frac{2}{3}} g(x) + \left(\frac{2}{3}a + c\right) f(x),\end{aligned}$$

Hence,

$$(\forall a, b, c \in \mathbb{R}) ax^2 + bx + c = a\frac{\sqrt{2}}{3} j(x) + b\sqrt{\frac{2}{3}} g(x) + \left(\frac{2}{3}a + c\right) f(x),$$

so we can see that S spans V. Hence, S is a basis of V.
We have now shown S is an orthonormal basis of V.

(c) Consider a finite dimensional vector space V over a field $\mathbb{F}$ with subspaces U and W.

i. What is meant by the sum $U + W$ of U and W ? [2 marks]

$U + W$ is the set of all possible linear combinations of vectors in U and W, i.e.

$$U + W = \{u + w : u \in U \wedge w \in W\}$$

ii. Show that $U + W = \text{span}(U \cup W)$. When is $U + W = U \cup W$? Justify your answer. [12 marks]

We start from the definition.

$$U \cup W = \{v \in V : v \in U \vee v \in W\}$$

Now, the span of $U \cup W$ is the set of all possible linear combinations of the form $u + w$, where $u \in U$ and $v \in V$.
Suppose that $v \in U + W$.

$$v \in U + W \Rightarrow (\exists u \in U)(\exists w \in W) v = u + w \Rightarrow v \in \text{span}(U \cup W).$$

Therefore, $U + W \subseteq \text{span}(U \cup W)$.
Suppose that $v \in \text{span}(U \cup W)$. Therefore, v is linear combination of vectors in either

U or W, i.e.

$$v \in \text{span}(U \cup W) \Rightarrow (\exists u_1, \ldots, u_I \in U)(\exists w_1, \ldots, w_J \in W) v = \sum_{i=1}^{I} \lambda_i u_i + \sum_{j=1}^{J} \mu_j w_i$$

Now, U and W are both vector spaces, so $\sum_{i=1}^{I} \lambda_i u_i = u \in U$ and $\sum_{j=1}^{J} \mu_j w_i = w \in W$. Therefore,

$$v \in \text{span}(U \cup W) \Rightarrow (\exists u \in U)(\exists w \in W) v = u + w \Rightarrow v \in U + W$$

Therefore, $\text{span}(U \cup W) \subseteq U + W$. We have $U + W \subseteq \text{span}(U \cup W)$ and $\text{span}(U \cup W) \subseteq U + W$, and so

$$U + W = \text{span}(U \cup W).$$

Now,

$$U + W = U \cup W \iff \text{span}(U \cup W) = U \cup W.$$

We can see that if $V = \mathbb{R}^2$ and U and W are non-parallel straight lines through the origin, then $\text{span}(U \cup W) \neq U \cup W$. However, if U is a subset of W or W is a subset of U, then $\text{span}(U \cup W) = U \cup W$. The converse is also true. We shall prove this.
First, suppose $U \subseteq W$. Then $U \cup W = W$ and so $\text{span}(U \cup W) = \text{span}(W)$. W is a vector space and so $\text{span}(W) = W$. To summarize, we have

$$U \subseteq W \Rightarrow \text{span}(U \cup W) = U \cup W.$$

Similarly,

$$W \subseteq U \Rightarrow \text{span}(U \cup W) = U \cup W.$$

We now want to show that

$$\text{span}(U \cup W) = U \cup W \Rightarrow U \subseteq W \vee W \subseteq U.$$

To prove the above statement, we shall prove its contrapositive

$$U \nsubseteq W \wedge W \nsubseteq U \Rightarrow \text{span}(U \cup W) \neq U \cup W$$

$$U \nsubseteq W \wedge W \nsubseteq U \Rightarrow ((\exists u \in V) u \in U \wedge u \notin W) \wedge (\exists w \in V) w \notin U \wedge w \in W)$$

Now we show $u + w \notin U \cup W$.

$$u + w \in U \cup W \Rightarrow u + w \in U \vee u + w \in W.$$

If $u + w \in U$ then (using the fact U is a vector space) $u + w + (-u) \in U \iff w \in U$, which is a contradiction. Similarly, $u + w \in W$ leads to a contradiction. Therefore, $u + w \notin U \cup W$, and so $\text{span}(U \cup W) \neq U \cup W$.

End of Paper